# Merged Solutions: Exercises 1.1–1.3, 1.5–1.7, 2.1–2.8, 3.1–3.4

July 29, 2026

## Exercise 1.1

(a) Show that the inner product has the following properties for all $x, y, z \in \mathbb{R}^n$ and $a \in \mathbb{R}$:

$$\langle x, y\rangle = \langle y, x\rangle, \qquad \langle x+y, z\rangle = \langle x, z\rangle + \langle y, z\rangle, \qquad \langle ax, y\rangle = a\langle x, y\rangle.$$

*Solution.* These follow from the Euclidean inner product $\langle x, y\rangle = \sum_{i=1}^n x_i y_i$.

(b) For $t \in \mathbb{R}$ and $x, y \in \mathbb{R}^n$ show

$$\|x+ty\|^2 = \|x\|^2 + 2t\langle x, y\rangle + t^2\|y\|^2 \geq 0.$$

*Solution.* Expand $\langle x+ty, x+ty\rangle$.

(c) Deduce Cauchy–Schwarz: $|\langle x, y\rangle| \leq \|x\|\,\|y\|$.
*Solution.* View $\|x+ty\|^2$ as a quadratic in $t$ and use non-positivity of its discriminant. Equality iff $x, y$ are linearly dependent.

(d) Deduce the triangle inequality $\|x+y\| \leq \|x\| + \|y\|$.

(e) Show the reverse triangle inequality $|\|x\| - \|y\|| \leq \|x-y\|$.

(f) Suppose $x = (x^1, \dots, x^n)^t$.

(i) $\max_k |x^k| \leq \|x\|$.

(ii) $\|x\| \leq \sqrt{n}\max_k |x^k|$.

*Solution.* Immediate from $\|x\|^2 = \sum (x^i)^2$.

## Exercise 1.2

Let $(x_i)$ and $(y_i)$ in $\mathbb{R}^n$ satisfy $x_i \to x$, $y_i \to y$.

(a) $x_i + y_i \to x + y$. *Proof.* $\|(x_i+y_i) - (x+y)\| \leq \|x_i - x\| + \|y_i - y\| \to 0$.

(b) $\langle x_i, y_i\rangle \to \langle x, y\rangle$; in particular $\|x_i\| \to \|x\|$.
*Proof.* Expand $\langle x_i, y_i\rangle - \langle x, y\rangle$ and use Cauchy–Schwarz.

(c) If $a_i \to a$ in $\mathbb{R}$, then $a_i x_i \to ax$.
*Proof.* $a_i x_i - ax = (a_i - a)(x_i - x) + (a_i - a)x + a(x_i - x)$.

## Exercise 1.3

Which of the following subsets of $\mathbb{R}^n$ is open?

(a) $\mathbb{R}^n$: **Yes**.

(b) $\varnothing$: **Yes**.

(c) $\{x \in \mathbb{R}^n : x^1 > 0\}$: **Yes**.

(d) $\{x \in \mathbb{R}^n : x^i \in [0,1]\}$: **No**.

(e) $\mathbb{Q}^n$: **No**.

## Exercise 1.5

(a) If $x_i \to x$ and $\|x_i\| < r$ for all $i$, then $\|x\| \leq r$.
*Proof.* Contradict $\|x\| > r$ using $\varepsilon = (\|x\| - r)/2$ and the reverse triangle inequality.

(b) The closure of $B_r(y) = \{x : \|x - y\| < r\}$ is $\overline{B_r(y)} = \{x : \|x - y\| \leq r\}$.
*Proof.* Use (a) for the first inclusion; for boundary points take $x_i = y + (1 - \frac{1}{i})(x - y)$.

## Exercise 1.6

Find $A^\circ$, $\overline{A}$ and $\partial A$.

(a) $A = \mathbb{R}^n$: $A^\circ = \mathbb{R}^n$, $\overline{A} = \mathbb{R}^n$, $\partial A = \varnothing$.

(b) $A = \{0\}$: $A^\circ = \varnothing$, $\overline{A} = \{0\}$, $\partial A = \{0\}$.

(c) $A = \{x : x^1 \geq 0\}$: $A^\circ = \{x : x^1 > 0\}$, $\overline{A} = A$, $\partial A = \{x : x^1 = 0\}$.

(d) $A = \{x : x^i \in [0,1] \ \forall i\}$: $A^\circ = \{x : x^i \in (0,1) \ \forall i\}$, $\overline{A} = A$, $\partial A = A \setminus A^\circ$.

(e) $A = \{x : x^i \in \mathbb{Q} \ \forall i\}$: $A^\circ = \varnothing$, $\overline{A} = \mathbb{R}^n$, $\partial A = \mathbb{R}^n$.

## Exercise 1.7

(a) If $U_1, U_2$ are open, then $U_1 \cup U_2$ and $U_1 \cap U_2$ are open.

(b) If $E_1, E_2$ are closed, then $E_1 \cup E_2$ and $E_1 \cap E_2$ are closed.

(c) For any collection $\mathcal{U}$ of open sets: $\bigcup \mathcal{U}$ is open; $\bigcap \mathcal{U}$ need not be open (e.g. $\cap_{n \geq 1}(-1/n, 1/n) = \{0\}$). For closed sets: arbitrary intersections are closed, finite unions are closed, infinite unions need not be closed.

## Exercise 2.1

A sequence $(x_i) \subset \mathbb{R}^n$ is Cauchy if $\forall \varepsilon > 0 \, \exists N : \ i, j \geq N \Rightarrow \|x_i - x_j\| < \varepsilon$.

(a) If $x_i \to x$, then $(x_i)$ is Cauchy. *Proof.* Triangle inequality with $\varepsilon/2$.

(b) If $(x_i)$ is Cauchy, then it is bounded. *Proof.* Take $\varepsilon = 1$.

(c) Every Cauchy sequence in $\mathbb{R}^n$ converges. *Proof.* Bounded $\Rightarrow$ has a convergent subsequence; use Cauchy property to pass to the full sequence.

**Remark.** In $\mathbb{R}$, $\mathbb{R}^n$, and $\mathbb{C} \cong \mathbb{R}^2$, a sequence is convergent iff it is Cauchy (completeness). In incomplete spaces (e.g. $\mathbb{Q}$), Cauchy does not imply convergent.

## Exercise 2.2

Suppose $A \subset \mathbb{R}^n$ and $f : A \to \mathbb{R}^m$. Then

$$\lim_{x\to p} f(x) = F \iff \text{ for every sequence } x_i \in A,\ x_i \neq p,\ x_i \to p, \text{ we have } f(x_i) \to F.$$

*Proof.* The forward direction is immediate from the $\varepsilon$–$\delta$ definition; the reverse is proved by contradiction constructing a sequence $x_k \to p$ with $f(x_k)$ staying $\varepsilon_0$ away from $F$.

## Exercise 2.3

(a) $f : \mathbb{R} \to \mathbb{R}^n$, $x \mapsto (x, 0, \dots, 0)^t$ is continuous: $\|f(x) - f(x_0)\| = |x - x_0|$.

(b) For $A \subset \mathbb{R}^n$, $f = (f^1, \dots, f^m) : A \to \mathbb{R}^m$ is continuous at $p$ iff each $f^k$ is continuous at $p$.

(c) Any polynomial map $f : \mathbb{R}^n \to \mathbb{R}$ is continuous (built from projections via sums/products).

## Exercise 2.4

Let $E \subset \mathbb{R}^n$ be compact and $f : E \to \mathbb{R}^m$ continuous.

(a) $f(E)$ is bounded. *Proof.* Each component $f^j$ attains its max/min on $E$.

(b) $f(E)$ is closed, hence compact. *Proof.* Use sequential closedness via compactness of $E$.

(c) If $E$ is merely closed, $f(E)$ need not be closed; e.g. $f(x) = \arctan x$ on $E = \mathbb{R}$.

## Exercise 2.5 (*)

(a) If $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous and $U$ open in $\mathbb{R}^m$, then $f^{-1}(U)$ is open.

(b) If $f^{-1}(U)$ is open for every open $U \subset \mathbb{R}^m$, then $f$ is continuous on $\mathbb{R}^n$.

## Exercise 2.6

Assume $f : \mathbb{R} \to \mathbb{R}$ differentiable with $|f'(x)| \leq M$ for all $x$.

(a) $|f(x) - f(y)| \leq M|x - y|$ by MVT.

(b) Hence $f$ is uniformly continuous (Lipschitz with constant $M$).

## Exercise 2.7

Which functions are (i) continuous, (ii) uniformly continuous on their domains?

(a) $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto e^x$: continuous, not uniformly continuous.

(b) $f : (0, 1) \to \mathbb{R}$, $x \mapsto e^x$: continuous, uniformly continuous (MVT and $e^c \leq e$).

(c) $f : \mathbb{R} \to \mathbb{R}$, $x \mapsto \sin x$: continuous, uniformly continuous ($|\sin x - \sin y| \leq |x - y|$).

(d) $f : [0, \infty) \to \mathbb{R}$, $x \mapsto \sqrt{x}$: continuous, uniformly continuous ($|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$).

*f)* $f : \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^n$, $x \mapsto x/\|x\|$: continuous, not uniformly continuous (fails near 0).

## Exercise 2.8

Let $A \subset \mathbb{R}^n$ and $f, g : A \to \mathbb{R}$.

(a) Suppose $f$ and $g$ are uniformly continuous and bounded on $A$. Then $fg$ is uniformly continuous and bounded on $A$.

*Solution (boundedness).* If $|f| \le M_f$ and $|g| \le M_g$ on $A$, then $|fg| \le M_f M_g$.

*Solution (uniform continuity).* Fix $\varepsilon > 0$ and set $\eta = \min\Big(1, \frac{\varepsilon}{2(M_f+M_g+1)}\Big)$. By uniform continuity of $f$ and $g$, there exists $\delta > 0$ such that $\|x - y\| < \delta$ implies $|f(x) - f(y)| < \eta$ and $|g(x) - g(y)| < \eta$. Then for such $x, y$,

$$\begin{aligned} |f(x)g(x) - f(y)g(y)| &= |[f(x) - f(y)][g(x) - g(y)] + g(y)[f(x) - f(y)] + f(y)[g(x) - g(y)]| \\ &\le \eta^2 + M_g\,\eta + M_f\,\eta \ \le\ (M_f + M_g + 1)\eta \ \le\ \varepsilon. \end{aligned}$$

Hence $fg$ is uniformly continuous.

(b) If $f$ and $g$ are uniformly continuous but not necessarily bounded, $fg$ need not be uniformly continuous.

*Counterexample.* On $\mathbb{R}$, take $f(x) = x$ and $g(x) = \sin x$. Both are uniformly continuous (Lipschitz). Assume $x \sin x$ were uniformly continuous. For $\varepsilon = 1$, let $\delta > 0$ be its modulus. Set $t = \min(\delta/2, 1)$ and take $y_k = k\pi$, $x_k = k\pi + t$. Then $|x_k - y_k| = t < \delta$ but

$$|x_k \sin x_k - y_k \sin y_k| = |(k\pi + t)\sin t| \ge k\pi \sin t \xrightarrow{k\to\infty} \infty,$$

contradicting uniform continuity. Thus $x \sin x$ is not uniformly continuous.

## Exercise 3.1

For $i \in \mathbb{N}$, define $f_i : [0, 1] \to \mathbb{R}$ by

$$f_i(x) = \begin{cases} 2^i x, & 0 \le x < 2^{-i}, \\ 2 - 2^i x, & 2^{-i} \le x < 2^{-(i-1)}, \\ 0, & x \ge 2^{-(i-1)}. \end{cases}$$

(a) *Sketch.* $f_i$ is a triangular "spike" supported on $[0, 2^{-(i-1)}]$: it rises linearly from 0 at $x = 0$ to height 1 at $x = 2^{-i}$, then decreases linearly back to 0 at $x = 2^{-(i-1)}$; afterwards it is identically 0. In particular, $\sup\limits_{x\in[0,1]} f_i(x) = 1$ for every $i$.

(b) *Pointwise limit.* For any fixed $x > 0$ choose $I$ such that $2^{-(i-1)} < x$ for all $i \ge I$; then $f_i(x) = 0$ for $i \ge I$. Also $f_i(0) = 0$ for all $i$. Hence $f_i(x) \to 0$ for every $x \in [0, 1]$.

(c) *Interchange of* lim *and* sup. We have

$$\lim_{i\to\infty} \sup_{x\in[0,1]} f_i(x) = \lim_{i\to\infty} 1 = 1, \qquad \text{while} \qquad \sup_{x\in[0,1]} \lim_{i\to\infty} f_i(x) = \sup_{x\in[0,1]} 0 = 0.$$

Therefore the equality

$$\lim_{i\to\infty} \sup_{x\in[0,1]} f_i(x) = \sup_{x\in[0,1]} \lim_{i\to\infty} f_i(x)$$

is *false* for this family. (This also shows $\{f_i\}$ does not converge uniformly to 0.)

## Exercise 3.2

State whether the sequence $(f_i)_{i=0}^{\infty}$ of functions converges pointwise and/or uniformly.

(a) $f_i : \mathbb{R} \to \mathbb{R}, \quad f_i(x) = e^x + \dfrac{1}{i+1}.$

*Solution.* For each $x$, $f_i(x) \to e^x$. Moreover

$$\sup_{x\in\mathbb{R}} |f_i(x) - e^x| = \sup_{x\in\mathbb{R}} \frac{1}{i+1} = \frac{1}{i+1} \xrightarrow{i\to\infty} 0,$$

so $f_i \to e^x$ *uniformly* on $\mathbb{R}$.

(b) $f_i : \mathbb{R} \to \mathbb{R}, \quad f_i(x) = x + 2^{-i}x^2.$

*Solution.* For fixed $x$, $f_i(x) \to x$. However

$$\sup_{x\in\mathbb{R}} |f_i(x) - x| = \sup_{x\in\mathbb{R}} 2^{-i}x^2 = +\infty \quad \text{for every fixed } i,$$

hence the convergence is *not* uniform on $\mathbb{R}$. (Nonetheless, on any bounded interval $[-M, M]$ the convergence is uniform since $\sup_{|x|\le M} |f_i(x) - x| = 2^{-i}M^2 \to 0$.)

(c) $f_i : [0, 1] \to \mathbb{R}$ given by

$$f_i(x) = \begin{cases} 2^i x, & 0 \le x < 2^{-i}, \\ 2 - 2^i x, & 2^{-i} \le x < 2^{-(i-1)}, \\ 0, & x \ge 2^{-(i-1)}. \end{cases}$$

*Solution.* For every $x \in [0, 1]$, $f_i(x) \to 0$ (as in Exercise 3.1). But $\sup_{x\in[0,1]} f_i(x) = 1$ for all $i$, so the convergence is *not* uniform on $[0, 1]$. (On $[\delta, 1]$ with $\delta > 0$, the convergence is uniform.)

## Exercise 3.3

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^{\infty}$ with $f_i : A \to \mathbb{R}^m$. Classify whether each statement is equivalent to (i) pointwise convergence $f_i \to f$, (ii) uniform convergence $f_i \to f$ uniformly, or (iii) neither.

(a) Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $i \ge N$:

$$\sup_{x\in A} \|f_i(x) - f(x)\| < \varepsilon.$$

*Classification: (ii) uniform convergence.* This is exactly the definition via the sup norm.

(b) For all $x \in A$, for all $i \in \mathbb{N}$ there exists $\varepsilon > 0$ such that

$$\|f_i(x) - f(x)\| < \varepsilon.$$

*Classification: (iii) neither.* The statement is always true (take, e.g., $\varepsilon = \|f_i(x) - f(x)\| + 1$), so it does not characterize convergence.

(c) There exists $N \in \mathbb{N}$ such that for all $x \in A$, $i \ge N$, $\varepsilon > 0$:

$$\|f_i(x) - f(x)\| < \varepsilon.$$

*Classification: (iii) neither.* Since the left-hand side is nonnegative, the inequality for *all* $\varepsilon > 0$ forces $\|f_i(x) - f(x)\| = 0$; i.e. there is $N$ with $f_i \equiv f$ on $A$ for all $i \ge N$. This condition is much stronger than uniform convergence and is not equivalent to it in general.

*d) $\forall \varepsilon > 0\ \forall x \in A\ \exists N \in \mathbb{N}\ \forall i \geq N: \ \|f_i(x) - f(x)\| < \varepsilon$.
*Classification: (i) pointwise convergence.* The quantifier order matches the definition; $N$ may depend on $x$ and $\varepsilon$.

*e) $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall x \in A\ \forall i \geq N: \ \|f_i(x) - f(x)\| < \varepsilon$.
*Classification: (ii) uniform convergence.* Here $N$ works simultaneously for all $x \in A$.

*f) $\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall x \in A: \ (\, i \geq N \ \Leftrightarrow\ \|f_i(x) - f(x)\| < \varepsilon\,)$.
*Classification: (iii) neither.* This is far stronger than uniform convergence (it also demands that for $i < N$ the error is $\geq \varepsilon$ for every $x$), and it need not hold even when $f_i \to f$ uniformly.

## Exercise 3.4

Suppose $A \subset \mathbb{R}^n$ and let $(f_i)_{i=0}^{\infty}$ with $f_i : A \to \mathbb{R}^m$ be a sequence of functions. Assume $f_i \to f$ *uniformly* on $A$. Show that if $B \subset A$, then $f_i \to f$ uniformly on $B$.

*Solution.* By uniform convergence on $A$, for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $i \geq N$ and all $x \in A$ we have $\|f_i(x) - f(x)\| < \varepsilon$. Since $B \subset A$, this inequality holds for all $x \in B$ as well, with the *same* $N$. Hence $f_i \to f$ uniformly on $B$.

Equivalently,

$$\sup_{x \in B} \|f_i(x) - f(x)\| \leq \sup_{x \in A} \|f_i(x) - f(x)\| \xrightarrow{i \to \infty} 0.$$

## Exercise 3.5 (*)

Let $E \subset \mathbb{R}^n$ be compact. Define $C^0(E, \mathbb{R}^m)$ to be the set of all continuous functions $f : E \to \mathbb{R}^m$, equipped with the norm

$$\|f\|_{C^0} := \sup_{x \in E} \|f(x)\| \in [0, \infty).$$

Show that $C^0(E, \mathbb{R}^m)$ is complete.

*Solution.* Let $(f_k)$ be a Cauchy sequence in the sup norm $\|\cdot\|_{C^0}$. Then, for every $x \in E$,

$$\|f_k(x) - f_\ell(x)\| \leq \|f_k - f_\ell\|_{C^0} \xrightarrow[k,\ell \to \infty]{} 0,$$

so for each $x$ the sequence $(f_k(x))$ is Cauchy in $\mathbb{R}^m$ and hence converges. Define $f(x) := \lim_{k \to \infty} f_k(x)$; this gives a function $f : E \to \mathbb{R}^m$.

*Uniform convergence.* Given $\varepsilon > 0$, choose $N$ so that $\|f_k - f_\ell\|_{C^0} < \varepsilon$ for all $k, \ell \geq N$. Fix $\ell \geq N$ and let $k \to \infty$; by the pointwise limit we obtain

$$\|f - f_\ell\|_{C^0} = \sup_{x \in E} \|f(x) - f_\ell(x)\| \leq \varepsilon.$$

Thus $f_k \to f$ uniformly on $E$. In particular, $\|f\|_{C^0} \leq \|f - f_\ell\|_{C^0} + \|f_\ell\|_{C^0} < \infty$, so $f$ is bounded.

*Continuity of the limit.* Let $x \in E$ and $\varepsilon > 0$. Choose $\ell \geq N$ so that $\|f - f_\ell\|_{C^0} < \varepsilon/3$. Since $f_\ell$ is continuous at $x$, there exists $\delta > 0$ with $\|y - x\| < \delta \Rightarrow \|f_\ell(y) - f_\ell(x)\| < \varepsilon/3$. Then for $\|y - x\| < \delta$,

$$\|f(y) - f(x)\| \leq \|f(y) - f_\ell(y)\| + \|f_\ell(y) - f_\ell(x)\| + \|f_\ell(x) - f(x)\| < \varepsilon.$$

Hence $f$ is continuous on $E$, i.e. $f \in C^0(E, \mathbb{R}^m)$.

*Convergence in norm.* With the same $N$ as above, fix $\ell \geq N$. For $k \geq N$,

$$\|f_k - f\|_{C^0} \leq \|f_k - f_\ell\|_{C^0} + \|f_\ell - f\|_{C^0} \leq \varepsilon + \varepsilon = 2\varepsilon,$$

so letting $\varepsilon \downarrow 0$ (or starting with $\varepsilon/2$) gives $\|f_k - f\|_{C^0} \to 0$.

Therefore every Cauchy sequence in $C^0(E, \mathbb{R}^m)$ converges (in $\|\cdot\|_{C^0}$) to an element of $C^0(E, \mathbb{R}^m)$. Thus $C^0(E, \mathbb{R}^m)$ is complete.

## Exercise 3.6 (*)

Suppose $f : (a, b) \to \mathbb{R}$ is uniformly continuous, and $(x_j)_{j=0}^\infty, (y_j)_{j=0}^\infty \subset (a, b)$ satisfy $x_j \to a$ and $y_j \to b$.

(a) Show that the sequences $(f(x_j))_{j=0}^\infty$ and $(f(y_j))_{j=0}^\infty$ are Cauchy.

*Solution.* Given $\varepsilon > 0$ choose $\delta > 0$ such that $|u - v| < \delta \Rightarrow |f(u) - f(v)| < \varepsilon$ (uniform continuity). Since $x_j \to a$ and $y_j \to b$, both $(x_j)$ and $(y_j)$ are Cauchy. Hence there is $N$ with $|x_j - x_k| < \delta$ and $|y_j - y_k| < \delta$ for all $j, k \geq N$, which implies $|f(x_j) - f(x_k)| < \varepsilon$ and $|f(y_j) - f(y_k)| < \varepsilon$ for $j, k \geq N$. Thus both image sequences are Cauchy.

(b) Setting $f_L := \lim_{j\to\infty} f(x_j)$ and $f_R := \lim_{j\to\infty} f(y_j)$, show that

$$\tilde{f} : [a, b] \to \mathbb{R}, \qquad \tilde{f}(x) = \begin{cases} f_L, & x = a, \\ f(x), & a < x < b, \\ f_R, & x = b, \end{cases}$$

is continuous on $[a, b]$.

*Solution.* By (a) the limits $f_L, f_R$ exist (Cauchy in $\mathbb{R}$ hence convergent). On $(a, b)$, $\tilde{f} = f$, so continuity holds. We check continuity at $a$; the case $b$ is analogous.

Let $\varepsilon > 0$. Pick $\delta > 0$ for uniform continuity of $f$ and choose $J$ large so that $|x_J - a| < \delta/2$ and $|f(x_J) - f_L| < \varepsilon$. If $0 < |t - a| < \delta/2$, then $|t - x_J| \leq |t - a| + |x_J - a| < \delta$, hence

$$|\tilde{f}(t) - \tilde{f}(a)| = |f(t) - f_L| \leq |f(t) - f(x_J)| + |f(x_J) - f_L| < \varepsilon + \varepsilon = 2\varepsilon.$$

Therefore $\tilde{f}$ is continuous at $a$. A symmetric argument using $(y_j)$ gives continuity at $b$. Thus $\tilde{f}$ is continuous on $[a, b]$.

(c) Deduce that $f$ is uniformly continuous on $(a, b)$ if and only if there exists a continuous $\tilde{f} : [a, b] \to \mathbb{R}$ with $\tilde{f}(x) = f(x)$ for all $x \in (a, b)$ (i.e. $\tilde{f}$ is an extension of $f$).

*Solution.* ($\Rightarrow$) If $f$ is uniformly continuous, parts (a)–(b) construct a continuous extension $\tilde{f}$ on $[a, b]$.

($\Leftarrow$) If such $\tilde{f}$ exists, then $\tilde{f}$ is continuous on the compact set $[a, b]$, hence uniformly continuous. Its restriction to $(a, b)$ equals $f$ and is therefore uniformly continuous.

## Exercise 3.8 (*)

Let $h : \mathbb{R} \to \mathbb{R}$ be the 1-periodic function satisfying $h(x) = |x|$ for $x \in [-\frac{1}{2}, \frac{1}{2}]$ (and $h(x + 1) = h(x)$). Define the Takagi function

$$T(x) = \sum_{i=0}^\infty \frac{1}{2^i} h(2^i x).$$

(a) *Sketch h.* On $[-\frac{1}{2}, \frac{1}{2}]$ it is the $V$–shape $|x|$ (cusp at 0), extended periodically with period 1. Equivalently,

$$h(x) = \operatorname{dist}(x, \mathbb{Z}) = \min_{k\in\mathbb{Z}} |x - k|,$$

so $0 \le h(x) \le \frac{1}{2}$, attaining 0 at integers and $\frac{1}{2}$ at half–integers.

(b) *Uniform continuity and differentiability.* Since $x \mapsto \operatorname{dist}(x, \mathbb{Z})$ is 1–Lipschitz, $h$ is uniformly continuous on $\mathbb{R}$. On each interval $(k, k + \frac{1}{2})$ and $(k - \frac{1}{2}, k)$ the function is affine with slope $-1$ or $+1$, hence differentiable there. It fails to be differentiable precisely at points congruent to 0 or $\frac{1}{2}$ modulo 1, i.e. at those $x$ with $2x \in \mathbb{Z}$. Thus $h$ is differentiable at every $x$ with $2x \notin \mathbb{Z}$.

(c) *Uniform convergence of the series and uniform continuity of $T$.* Since $0 \le h \le \frac{1}{2}$,

$$\left|\frac{1}{2^i} h(2^i x)\right| \le \frac{1}{2^{i+1}} \qquad (\forall x \in \mathbb{R}),$$

and $\sum_{i=0}^{\infty} 2^{-(i+1)} < \infty$. By the Weierstrass $M$–test, the series converges uniformly on $\mathbb{R}$ to a continuous function $T$. Each summand $x \mapsto 2^{-i}h(2^i x)$ is uniformly continuous (indeed 1–Lipschitz), hence $T$, being a uniform limit of uniformly continuous functions, is uniformly continuous on $\mathbb{R}$.

(d) Let $x \in \mathbb{R}$. There exist unique $u_n, v_n$ such that

(i) $2^n u_n$, $2^n v_n \in \mathbb{Z}$,

(ii) $v_n - u_n = 2^{-n}$,

(iii) $u_n \le x < v_n$.

*Solution.* Take $u_n = \frac{\lfloor 2^n x\rfloor}{2^n}$ and $v_n = u_n + 2^{-n}$. Then (i)–(iii) hold. Uniqueness: if $u_n' \le x < v_n'$ with (i)–(ii), then $2^n u_n' \le 2^n x < 2^n v_n'$ and $2^n v_n' - 2^n u_n' = 1$, so $2^n u_n' = \lfloor 2^n x\rfloor = 2^n u_n$, hence $u_n' = u_n$ and $v_n' = v_n$.

(e) For $i \ge 0$ set $h_i(x) = 2^{-i}h(2^i x)$. Show that:

(i) $h_i(u_n) = h_i(v_n) = 0$ for $i \ge n$.

(ii) $\dfrac{h_i(v_n) - h_i(u_n)}{v_n - u_n} = \pm 1$ for $i < n$.

*Solution.* (i) Since $2^n u_n, 2^n v_n \in \mathbb{Z}$, for $i \ge n$ also $2^i u_n, 2^i v_n \in \mathbb{Z}$, hence $h(2^i u_n) = h(2^i v_n) = 0$.

(ii) The map $x \mapsto 2^i x$ sends $[u_n, v_n]$ onto the interval $\left[\frac{K}{2^{n-i}}, \frac{K+1}{2^{n-i}}\right]$ where $K = \lfloor 2^n x\rfloor$. Its length is $2^i(v_n - u_n) = 2^{-(n-i)} \le \frac{1}{2}$, so it contains no half–integer. Hence $h$ is affine with slope $\pm 1$ on this interval, and therefore

$$\frac{h_i(v_n) - h_i(u_n)}{v_n - u_n} = \frac{2^{-i}\big(h(2^i v_n) - h(2^i u_n)\big)}{2^{-n}} = 2^{n-i} \cdot 2^{-i} \cdot (\pm 2^{-(n-i)}) = \pm 1.$$

(f) *Conclude that*

$$\frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{i=0}^{n-1} \frac{h_i(v_n) - h_i(u_n)}{v_n - u_n},$$

*and that the sum on the right does not converge as $n \to \infty$.*

*Solution.* Since $h_i(u_n) = h_i(v_n) = 0$ for $i \geq n$,

$$T(v_n) - T(u_n) = \sum_{i=0}^{\infty} \big(h_i(v_n) - h_i(u_n)\big) = \sum_{i=0}^{n-1} \big(h_i(v_n) - h_i(u_n)\big).$$

Divide by $v_n - u_n$ to get the displayed identity. By (e)(ii), each term of the sum equals $\pm 1$, so the sequence of partial sums cannot converge (successive differences are $\pm 1 \not\to 0$). Hence the right–hand side does not converge as $n \to \infty$. In particular, $T$ is nowhere differentiable.

## Exercise 3.8 (g)

Show that the Takagi function $T(x) = \sum_{i=0}^{\infty} 2^{-i} h(2^i x)$ is not differentiable at any $x$.

*Solution.* Let $u_n = \frac{\lfloor 2^n x \rfloor}{2^n}$ and $v_n = u_n + 2^{-n}$. Then $u_n \uparrow x$, $v_n \downarrow x$, and

$$\frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{i=0}^{n-1} \frac{h_i(v_n) - h_i(u_n)}{v_n - u_n}, \qquad h_i(t) = 2^{-i} h(2^i t).$$

Each term equals $\pm 1$, so the partial sums do not converge. If $T'(x)$ existed, these difference quotients would converge to $T'(x)$, a contradiction. Hence $T$ is nowhere differentiable.

## Exercise 3.9 (*)

Suppose $F = (f_i)_{i=0}^{\infty}$ is a family of continuous functions $f_i : [a,b] \to \mathbb{R}$, each differentiable on $(a,b)$, and assume

$$\sup_{x \in (a,b)} |f_i'(x)| \leq M \qquad \text{for all } i,$$

for some $M$ independent of $i$. Show that $F$ is uniformly equicontinuous.

*Solution.* By the mean value theorem, for any $x, y \in [a,b]$ with $x \neq y$ there exists $c \in (x,y)$ such that

$$|f_i(x) - f_i(y)| = |f_i'(c)|\,|x-y| \leq M\,|x-y|, \qquad \text{for every } i.$$

Thus each $f_i$ is Lipschitz with the same constant $M$. Given $\varepsilon > 0$, choose $\delta = \varepsilon/M$. Then for all $i$ and all $x, y \in [a,b]$, if $|x-y| < \delta$ we have $|f_i(x) - f_i(y)| < \varepsilon$. Hence $F$ is uniformly equicontinuous on $[a,b]$.

## Definitions and Examples

**Supremum and Infimum.** For a set $S \subset \mathbb{R}$:

- The *supremum* (least upper bound) is the smallest $M \in \mathbb{R}$ such that $x \leq M$ for all $x \in S$. Denoted $\sup S$.
- The *infimum* (greatest lower bound) is the largest $m \in \mathbb{R}$ such that $m \leq x$ for all $x \in S$. Denoted $\inf S$.

**Example.** For $S = (0,1)$, $\sup S = 1$, $\inf S = 0$.

**Partitions, Upper and Lower Sums.** A *partition* of $[a, b]$ is a finite set

$$P = \{x_0, x_1, \ldots, x_n\}, \quad a = x_0 < x_1 < \cdots < x_n = b.$$

For a bounded function $f : [a, b] \to \mathbb{R}$:

$$U(f, P) = \sum_{i=1}^{n} M_i(x_i - x_{i-1}), \quad M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\},$$

$$L(f, P) = \sum_{i=1}^{n} m_i(x_i - x_{i-1}), \quad m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}.$$

**Example.** For $f(x) = x$ on $[0, 1]$ with $P = \{0, 0.5, 1\}$:

$$U(f, P) = 0.75, \quad L(f, P) = 0.25.$$

**Integrability.** A bounded $f : [a, b] \to \mathbb{R}$ is *Riemann integrable* if

$$\sup_P L(f, P) = \inf_P U(f, P).$$

**Example.** $f(x) = x$ on $[0, 1]$ is integrable with $\int_0^1 x dx = \frac{1}{2}$.

**Characteristic Function.** For $A \subset \mathbb{R}$, the *characteristic function* is

$$\chi_A(x) = \begin{cases} 1 & x \in A, \\ 0 & x \notin A. \end{cases}$$

**Example.** If $A = [0, \frac{1}{2}]$, then $\int_0^1 \chi_A(x) dx = \frac{1}{2}$.

# Solutions

## Exercise 4.1

*Solution.*

a) $\sup_{x \in A}(-f(x)) = -\inf_{x \in A} f(x)$.

b) $\inf_{x \in A}(-f(x)) = -\sup_{x \in A} f(x)$.

c) $\sup_{x \in A} \lambda f(x) = \lambda \sup_{x \in A} f(x)$ for $\lambda \geq 0$.

d) $\sup_{x \in A}(f(x) + g(x)) \leq \sup f + \sup g$.

e) $\inf_{x \in A}(f(x) + g(x)) \geq \inf f + \inf g$.

f) $\inf_{x \in A} \lambda f(x) = \lambda \inf f(x)$.

g) $|\sup f| \leq \sup |f|$.

h) $|\inf f| \leq \sup |f|$.

## Exercise 4.2

*Solution.*

a) $U(-f, P) = -L(f, P)$.

b) $L(-f, P) = -U(f, P)$.

c) $U(\lambda f, P) = \lambda U(f, P)$ for $\lambda \geq 0$.

d) $L(\lambda f, P) = \lambda L(f, P)$.

e) $U(f + g, P) \leq U(f, P) + U(g, P)$.

f) $L(f + g, P) \geq L(f, P) + L(g, P)$.

## Exercise 4.3

*Solution.* If $P \preceq Q$, then $U(f, Q) \leq U(f, P)$ and $L(f, Q) \geq L(f, P)$. Hence

$$0 \leq U(f, Q) - L(f, Q) \leq U(f, P) - L(f, P).$$

## Exercise 4.4

*Solution.* The function $\chi_A$ is Riemann integrable since it has only finitely many discontinuities. Moreover,

$$\int_{-1}^{1} \chi_A(x)\, dx = 0,$$

because a finite set has measure zero.

## Exercise 4.5

Suppose that $a < c < b$ and suppose $f : [a, b] \to \mathbb{R}$ is a bounded function. Let $\mathcal{P}$ be a partition of $[a, b]$ of the form:

$$\mathcal{P} = (a, x_1, \ldots, x_{l-1}, x_l = c, x_{l+1}, \ldots, x_{k-1}, b).$$

Define $\mathcal{P}_L$ and $\mathcal{P}_R$ to be partitions of $[a, c]$ and $[c, b]$ respectively:

$$\mathcal{P}_L = (a, x_1, \ldots, x_{l-1}, c), \qquad \mathcal{P}_R = (c, x_{l+1}, \ldots, x_{k-1}, b).$$

Show that

$$L(f, \mathcal{P}) = L(f|_{[a,c]}, \mathcal{P}_L) + L(f|_{[c,b]}, \mathcal{P}_R), \qquad U(f, \mathcal{P}) = U(f|_{[a,c]}, \mathcal{P}_L) + U(f|_{[c,b]}, \mathcal{P}_R).$$

*Solution.* By definition,

$$L(f, \mathcal{P}) = \sum_{i=1}^{k} m_i(x_i - x_{i-1}), \quad m_i = \inf_{x \in [x_{i-1}, x_i]} f(x).$$

The partition $\mathcal{P}$ splits at $c$: intervals before $c$ contribute the lower sum on $[a, c]$, those after $c$ contribute on $[c, b]$. Thus

$$L(f, \mathcal{P}) = L(f|_{[a,c]}, \mathcal{P}_L) + L(f|_{[c,b]}, \mathcal{P}_R).$$

The same argument with suprema shows

$$U(f, \mathcal{P}) = U(f|_{[a,c]}, \mathcal{P}_L) + U(f|_{[c,b]}, \mathcal{P}_R).$$

**Example.** For $f(x) = x$, $a = 0$, $b = 2$, $c = 1$ with partition $\mathcal{P} = (0, 1, 2)$:

$$L(f, \mathcal{P}_L) = 0, \quad L(f, \mathcal{P}_R) = 1, \quad U(f, \mathcal{P}_L) = 1, \quad U(f, \mathcal{P}_R) = 2.$$

So $L(f, \mathcal{P}) = 1$ and $U(f, \mathcal{P}) = 3$, which indeed split into left and right parts.

## Exercise 4.6

Suppose $f, g : [a, b] \to \mathbb{R}$ are integrable.

a) Show that

$$(b-a) \inf_{x \in [a,b]} f(x) \leq \int_a^b f(x)\, dx \leq (b-a) \sup_{x \in [a,b]} f(x).$$

b) Establish the estimate:

$$\left| \int_a^b f(x)\, dx \right| \leq (b-a) \sup_{x \in [a,b]} |f(x)|.$$

c) Show that if $0 \leq f(x)$ for all $x \in [a, b]$, then

$$0 \leq \int_a^b f(x)\, dx.$$

d) Show that if $f(x) \leq g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f(x)\, dx \leq \int_a^b g(x)\, dx.$$

e) Show that

$$\left| \int_a^b f(x)\, dx - \int_a^b g(x)\, dx \right| \leq \int_a^b |f(x) - g(x)|\, dx.$$

*Solution.*

a) Since $\inf f \leq f(x) \leq \sup f$ for all $x$, integrating and multiplying by $(b-a)$ gives

$$(b-a) \inf f \leq \int_a^b f(x)\, dx \leq (b-a) \sup f.$$

b) Apply part (a) to $f$ and to $-f$:

$$-(b-a) \sup |f| \leq \int_a^b f(x)\, dx \leq (b-a) \sup |f|.$$

Taking absolute values gives

$$\left| \int_a^b f(x)\, dx \right| \leq (b-a) \sup |f|.$$

c) If $f \geq 0$, then $\inf f \geq 0$. By part (a),

$$0 \leq \int_a^b f(x)\, dx.$$

d) If $f \le g$, set $h = g - f \ge 0$. Then by part (c),

$$0 \le \int_a^b h(x)\,dx = \int_a^b g(x)\,dx - \int_a^b f(x)\,dx.$$

e) By the triangle inequality for integrals:

$$\int_a^b (f-g)\,dx \le \int_a^b |f-g|\,dx, \qquad \int_a^b (g-f)\,dx \le \int_a^b |f-g|\,dx.$$

Thus

$$\left|\int_a^b (f-g)\,dx\right| \le \int_a^b |f-g|\,dx.$$

**Example.** For $f(x) = x$, $g(x) = x^2$ on $[0,1]$:

$$\int_0^1 f(x)\,dx = \tfrac{1}{2}, \quad \int_0^1 g(x)\,dx = \tfrac{1}{3}.$$

So

$$\left|\tfrac{1}{2} - \tfrac{1}{3}\right| = \tfrac{1}{6}, \quad \int_0^1 |x - x^2|\,dx = \tfrac{1}{6},$$

and equality holds.

### Exercise 4.7

Consider Thomae's function $f : [0,1] \to \mathbb{R}$:

$$f(x) = \begin{cases} 1, & x = 0, \\ \frac{1}{q}, & x = \frac{p}{q} \in \mathbb{Q},\ \gcd(p,q) = 1, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Show that $f$ is integrable.

*Solution.* First, $0 \le f(x) \le 1$ for all $x$, so $f$ is bounded.

**Continuity:** If $x$ is irrational, then $f(x) = 0$. For any $\varepsilon > 0$, rationals with denominator $q$ large enough satisfy $1/q < \varepsilon$, and they are isolated; hence in a small neighborhood of $x$ we avoid rationals with small denominator. Thus $f$ is continuous at all irrationals.

If $x = \frac{p}{q}$ rational, then $f(x) = 1/q > 0$, but arbitrarily close are irrationals with $f = 0$. So $f$ is discontinuous at every rational.

Therefore the discontinuity set is the rationals in $[0,1]$, a countable set.

**Integrability:** By a standard theorem, a bounded function on $[a,b]$ with discontinuities on a measure-zero set is Riemann integrable. Since $\mathbb{Q} \cap [0,1]$ has measure zero, $f$ is integrable.

**Integral value:** Since irrationals form full measure and $f(x) = 0$ there,

$$\int_0^1 f(x)\,dx = 0.$$

**Example intuition:** The graph shows spikes at rationals $p/q$ of height $1/q$, but these spikes shrink so fast that their total area vanishes.

## Exercise 5.1

Suppose $f : [a, b] \to \mathbb{R}$ is continuous, $f(x) \geq 0$, and $\int_a^b f(x)dx = 0$.

a) Show that $f(x) = 0$ on $[a, b]$.

b) Does the result hold without continuity?

*Solution.* (a) If $f(c) > 0$ for some $c$, by continuity there is $\delta > 0$ such that $f(x) > \frac{1}{2}f(c) > 0$ on $(c - \delta, c + \delta)$. Then

$$\int_a^b f(x)dx \geq \int_{c-\delta}^{c+\delta} f(x)dx > 0,$$

contradiction. Hence $f \equiv 0$.

(b) Without continuity the result fails. Example: the Dirichlet function $\chi_{\mathbb{Q}\cap[0,1]}$ has $\int_0^1 f = 0$, but $f(x) \neq 0$ for rationals.

## Exercise 5.2

Show $\int_0^1 x^2 dx = \frac{1}{3}$ using Riemann sums.

*Solution.* Partition $[0, 1]$ into $l$ equal subintervals. Right endpoint sum:

$$S_l = \frac{1}{l}\sum_{j=1}^{l}\left(\frac{j}{l}\right)^2 = \frac{1}{l^3}\sum_{j=1}^{l} j^2.$$

Since $\sum_{j=1}^{l} j^2 = \frac{l(l+1)(2l+1)}{6}$, we get

$$S_l = \frac{l(l+1)(2l+1)}{6l^3} \to \frac{1}{3} \quad (l \to \infty).$$

## Exercise 5.3

Show

$$\lim_{k\to\infty}\sum_{j=0}^{k-1}\frac{k}{j^2 + k^2} = \int_0^1 \frac{1}{1 + x^2}dx.$$

*Solution.* Rewrite

$$\frac{k}{j^2 + k^2} = \frac{1}{k} \cdot \frac{1}{1 + (j/k)^2}.$$

So sum is a Riemann sum for $\int_0^1 \frac{1}{1+x^2}dx$. Thus

$$\lim_{k\to\infty}\sum_{j=0}^{k-1}\frac{k}{j^2 + k^2} = \int_0^1 \frac{1}{1 + x^2}dx = \arctan(1) - \arctan(0) = \frac{\pi}{4}.$$

## Exercise 5.4

Define $f : [-1, 1] \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

a) Show $f$ is integrable and compute $F(x) = \int_{-1}^{x} f(t)dt$.

b) Show $F$ is not differentiable at 0.

*Solution.* (a) $f$ has one discontinuity at 0, hence integrable. For $x \le 0$, $F(x) = 0$. For $x > 0$,

$$F(x) = \int_0^x 1dt = x.$$

So $F(x) = 0$ for $x \le 0$, $F(x) = x$ for $x > 0$.

(b) For $x < 0$, $F'(x) = 0 = f(x)$. For $x > 0$, $F'(x) = 1 = f(x)$. At $x = 0$, left derivative $0 \neq 1$ right derivative, so $F$ is not differentiable at 0.

### Exercise 5.5

Suppose $f : [a, b] \to \mathbb{R}$ is integrable and $F : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on $(a, b)$, and satisfies

$$F'(x) = f(x) \quad \text{for all } x \in (a, b).$$

Show that:

1. For a partition $P = (x_0, \dots, x_k)$, there exists $t_j \in [x_j, x_{j+1}]$ such that

$$F(x_{j+1}) - F(x_j) = f(t_j)(x_{j+1} - x_j).$$

2. For a tagged partition $(P, \tau)$ with $\tau = (t_0, \dots, t_{k-1})$, one has

$$R(f, P, \tau) = F(b) - F(a).$$

3. Deduce that the continuity assumption in the integrability theorem (Corollary 2.14) can be weakened.

**Background.**

- **Mean Value Theorem (MVT).** If $F$ is continuous on $[u, v]$ and differentiable on $(u, v)$, then there exists $t \in (u, v)$ such that

$$F(v) - F(u) = F'(t)(v - u).$$

- **Riemann Sum.** For a tagged partition $(P, \tau)$, the Riemann sum is

$$R(f, P, \tau) = \sum_{j=0}^{k-1} f(t_j)(x_{j+1} - x_j).$$

*Solution.* **(a)** Apply the Mean Value Theorem to each interval $[x_j, x_{j+1}]$. Since $F$ is continuous on $[x_j, x_{j+1}]$ and differentiable on $(x_j, x_{j+1})$, there exists $t_j \in (x_j, x_{j+1})$ such that

$$F(x_{j+1}) - F(x_j) = F'(t_j)(x_{j+1} - x_j).$$

But $F'(t_j) = f(t_j)$, hence

$$F(x_{j+1}) - F(x_j) = f(t_j)(x_{j+1} - x_j).$$

**(b)** Summing over all subintervals:

$$\sum_{j=0}^{k-1}[F(x_{j+1}) - F(x_j)] = F(b) - F(a).$$

By part (a), each difference equals $f(t_j)(x_{j+1} - x_j)$, so

$$R(f, P, \tau) = \sum_{j=0}^{k-1} f(t_j)(x_{j+1} - x_j) = F(b) - F(a).$$

**(c)** Now consider any sequence of partitions $(P_n)$ of $[a, b]$ with mesh $\|P_n\| \to 0$. For each partition, choose tags as in (a). Then by part (b), each Riemann sum satisfies

$$R(f, P_n, \tau_n) = F(b) - F(a).$$

Thus all such Riemann sums converge to the same value $F(b) - F(a)$, independently of the sequence of partitions. Therefore $f$ is Riemann integrable and

$$\int_a^b f(x)\, dx = F(b) - F(a).$$

This shows that the requirement "$f$ is continuous" in Corollary 2.14 can be weakened: it suffices that $f$ be integrable and arise as the derivative of some differentiable function $F$. In other words, continuity of $f$ is not essential for the Fundamental Theorem of Calculus to hold.

### Exercise 5.6

a) For $0 < a < b$, show

$$\int_a^b \frac{1}{1+t^2}\, dt = \int_{1/b}^{1/a} \frac{1}{1+t^2}\, dt,$$

using the substitution $\phi(t) = 1/t$.

b) Setting $a = -1$, $b = 1$ in the above, one seems to obtain

$$\int_{-1}^{1} \frac{1}{1+x^2}\, dx = \int_{-1}^{1} \frac{1}{1+x^2}\, dx.$$

Explain what went wrong.

**Background.** The substitution rule requires the substitution $\phi$ to be differentiable and monotone on the interval. For $\phi(t) = 1/t$, we have $\phi'(t) = -1/t^2$.

*Solution.* **(a)** With $x = \phi(t) = 1/t$, we have $dx = -\frac{1}{t^2}dt$ and

$$1 + x^2 = 1 + \frac{1}{t^2} = \frac{1+t^2}{t^2}.$$

Thus

$$\frac{1}{1+x^2}\, dx = \frac{1}{\frac{1+t^2}{t^2}} \cdot \left(-\frac{1}{t^2}dt\right) = -\frac{1}{1+t^2}dt.$$

When $x = a$, $t = 1/a$; when $x = b$, $t = 1/b$. Therefore

$$\int_a^b \frac{1}{1+x^2}\, dx = \int_{1/a}^{1/b} -\frac{1}{1+t^2}\, dt = \int_{1/b}^{1/a} \frac{1}{1+t^2}\, dt.$$

**(b)** If $a = -1$, $b = 1$, then $\phi(t) = 1/t$ is not continuous or monotone on $[-1, 1]$, since it has a singularity at 0. The substitution theorem requires $\phi$ to be differentiable and monotone on the whole interval, which fails here. Thus the formula cannot be applied directly across $[-1, 1]$.

In fact, the integral is computed directly as

$$\int_{-1}^{1} \frac{1}{1 + x^2}\,dx = 2\arctan(1) = \frac{\pi}{2}.$$

If one wishes to use the substitution $x = 1/t$, the interval must be split at 0 and treated separately on $[-1, 0)$ and $(0, 1]$.

## Exercise 5.7

Let $I_1 = [\alpha, \beta]$, $I_2 = [\gamma, \delta]$ be finite intervals. Suppose that $\phi : I_1 \to I_2$ is continuous on $I_1$ and differentiable on $(\alpha, \beta)$ with $\phi'$ integrable on $I_1$. Suppose that $f : I_2 \to \mathbb{R}$ is continuous. Show that:

$$\int_{\phi(\alpha)}^{\phi(\beta)} f(t)\,dt = \int_{\alpha}^{\beta} f(\phi(t))\,\phi'(t)\,dt.$$

**Background.** This is the standard *change of variables formula* for Riemann integrals. It generalizes Exercise 5.6 and can be proven by combining:

- the Mean Value Theorem on each subinterval of a partition,
- the telescoping sum argument as in Exercise 5.5,
- the fact that Riemann sums converge to the integral when mesh $\to 0$.

*Solution.* Partition $I_1 = [\alpha, \beta]$ into $P = (x_0, \ldots, x_k)$. On each $[x_j, x_{j+1}]$, apply the Mean Value Theorem to $F(u) = \int_\gamma^u f(t)\,dt$, which is differentiable with derivative $F'(u) = f(u)$.

Then for each subinterval there exists $t_j \in (x_j, x_{j+1})$ such that

$$F(\phi(x_{j+1})) - F(\phi(x_j)) = f(\phi(t_j))\big(\phi(x_{j+1}) - \phi(x_j)\big).$$

Summing over $j$ gives a telescoping series:

$$F(\phi(\beta)) - F(\phi(\alpha)) = \sum_{j=0}^{k-1} f(\phi(t_j))\big(\phi(x_{j+1}) - \phi(x_j)\big).$$

But $\phi(x_{j+1}) - \phi(x_j) = \phi'(\xi_j)(x_{j+1} - x_j)$ for some $\xi_j \in (x_j, x_{j+1})$. Thus the sum is a Riemann sum for $\int_\alpha^\beta f(\phi(t))\phi'(t)\,dt$.

Passing to the limit as the mesh $\to 0$, we obtain

$$\int_{\phi(\alpha)}^{\phi(\beta)} f(t)\,dt = \int_{\alpha}^{\beta} f(\phi(t))\phi'(t)\,dt.$$

This is exactly the substitution formula.

## Exercise 5.8

By modifying the example of Section 2.3.1, show that there exists a family $(f_j)_{j=0}^{\infty}$ of integrable functions $f_j : [0, 1] \to \mathbb{R}$ such that:

$$f_j(x) \to 0 \quad \text{for all } x \in [0, 1],$$

while

$$\int_0^1 f_j(x)\,dx \;\to\; \infty.$$

**Background.** Pointwise convergence $f_j(x) \to 0$ does not imply convergence of integrals. A classic counterexample uses "spikes" of increasing height but decreasing width, so that the area (integral) diverges even though each fixed $x$ eventually lies outside all spikes.

*Solution.* Define

$$f_j(x) = \begin{cases} j^2, & 0 \le x \le \frac{1}{j}, \\ 0, & \frac{1}{j} < x \le 1. \end{cases}$$

**Step 1. Pointwise convergence.** Fix $x \in (0, 1]$. For sufficiently large $j$, we have $x > 1/j$, so $f_j(x) = 0$. Thus $f_j(x) \to 0$ for all $x \in (0, 1]$. Also $f_j(0) = j^2$, which diverges, but convergence is only required for each fixed $x \in [0, 1]$, and for $x = 0$, note $f_j(0)$ does not tend to 0. If we instead define $f_j(0) = 0$, the property holds for all $x$.

**Step 2. Integrals.** Compute:

$$\int_0^1 f_j(x)\,dx = \int_0^{1/j} j^2\,dx = j^2 \cdot \frac{1}{j} = j.$$

Hence $\int_0^1 f_j(x)\,dx \to \infty$ as $j \to \infty$.

**Conclusion.** This provides the required example: although $f_j(x) \to 0$ pointwise, the integrals diverge to infinity. It shows that pointwise convergence is not enough to pass limits under the integral sign.

## Exercise 6.1

Suppose $f : [0, 1) \to [0, 1)$ is nonnegative, continuous, and monotone decreasing. Let $N, M \in \mathbb{Z}$.

a) Show that

$$\sum_{n=N}^{M} f(n) \;\ge\; \int_N^{M+1} f(x)\,dx.$$

b) Show that

$$\sum_{n=N+1}^{M} f(n) \;\le\; \int_N^{M} f(x)\,dx.$$

c) Deduce

$$\int_N^{M+1} f(x)\,dx \;\le\; \sum_{n=N}^{M} f(n) \;\le\; f(N) + \int_N^{M} f(x)\,dx.$$

d) Show that $\sum_{n=N}^{\infty} f(n)$ converges iff the improper integral $\int_N^{\infty} f(x)\,dx$ exists.

e) Conclude that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

*Solution.* (a) On $[n, n+1]$, $f(x) \le f(n)$, so

$$\int_n^{n+1} f(x)\,dx \le f(n).$$

Summing yields

$$\int_N^{M+1} f(x)\,dx \le \sum_{n=N}^{M} f(n).$$

(b) On $[n, n+1]$, $f(x) \geq f(n+1)$, so

$$\int_n^{n+1} f(x)\,dx \geq f(n+1).$$

Summing gives

$$\int_N^M f(x)\,dx \geq \sum_{n=N+1}^{M} f(n).$$

(c) Combining (a) and (b):

$$\int_N^{M+1} f(x)\,dx \leq \sum_{n=N}^{M} f(n) \leq f(N) + \int_N^M f(x)\,dx.$$

(d) The error term between the sum and integral is at most $f(N)$, independent of $M$. Hence $\sum f(n)$ converges iff $\int f(x)$ converges.

(e) For $f(x) = 1/x$, $\int_1^M \frac{1}{x}\,dx = \log M \to \infty$. So the harmonic series diverges.

### Exercise 6.2

Suppose $f : [0, \infty) \to \mathbb{R}$ is nonnegative and decreasing. Show that

$$\sum_{n=1}^{\infty} f(n) < \infty \qquad \Longrightarrow \qquad \lim_{x\to\infty} x f(x) = 0.$$

*Solution.* Suppose $\sum f(n)$ converges. Then by the integral test, $\int_1^\infty f(x)\,dx$ also converges.

Assume for contradiction that $\limsup_{x\to\infty} x f(x) > \epsilon > 0$. Then there exists a sequence $x_k \to \infty$ with

$$f(x_k) \geq \frac{\epsilon}{x_k}.$$

For integers $n$ with $n \leq x_k < 2n$, monotonicity gives

$$f(n) \geq f(x_k) \geq \frac{\epsilon}{x_k} \geq \frac{\epsilon}{2n}.$$

Thus infinitely many terms satisfy $f(n) \geq \frac{\epsilon}{2n}$. Hence

$$\sum f(n) \;\geq\; \frac{\epsilon}{2} \sum \frac{1}{n},$$

which diverges. This contradicts convergence. Therefore

$$\lim_{x\to\infty} x f(x) = 0.$$

## Exercise 6.3

Suppose $f : [0, \infty) \to \mathbb{R}$ is nonnegative, monotone decreasing, and integrable on $[0, \infty)$. Show that

$$\lim_{x \to \infty} f(x) = 0.$$

*Solution.* Assume for contradiction that $\lim_{x \to \infty} f(x) = L > 0$. Then there exists $X$ such that $f(x) \geq L/2$ for all $x \geq X$. Hence

$$\int_X^\infty f(x)\, dx \;\geq\; \int_X^\infty \frac{L}{2}\, dx = \infty,$$

contradiction. Therefore $L = 0$, i.e.

$$\lim_{x \to \infty} f(x) = 0.$$

## Exercise 6.4

Suppose $(a_n)_{n=1}^\infty$ is a sequence of positive decreasing real numbers. Show that

$$\sum_{n=1}^\infty a_n \quad \text{converges} \quad \iff \quad \sum_{n=1}^\infty 2^n a_{2^n} \text{ converges.}$$

*Solution.* Partition $\sum a_n$ into dyadic blocks:

$$S_n = \sum_{k=2^n}^{2^{n+1}-1} a_k.$$

Since $a_n$ is decreasing, for $k \in [2^n, 2^{n+1} - 1]$,

$$a_{2^{n+1}} \leq a_k \leq a_{2^n}.$$

There are $2^n$ terms, so

$$2^n a_{2^{n+1}} \leq S_n \leq 2^n a_{2^n}.$$

Summing over $n = 0, 1, \ldots, m$:

$$\sum_{n=0}^m 2^n a_{2^{n+1}} \;\leq\; \sum_{k=1}^{2^{m+1}-1} a_k \;\leq\; \sum_{n=0}^m 2^n a_{2^n}.$$

Thus $\sum a_n$ converges iff $\sum 2^n a_{2^n}$ converges.

## Exercise 6.5

Suppose $f : [0, \infty) \to \mathbb{R}$ is nonnegative and decreasing. Show that

$$\sum_{n=1}^\infty f(n) \quad \text{converges} \quad \iff \quad \sum_{k=0}^\infty 2^k f(2^k) \text{ converges.}$$

*Solution.* Let $a_n = f(n)$. Then $(a_n)$ is decreasing and nonnegative. By the Cauchy Condensation Test (Exercise 6.4),

$$\sum_{n=1}^{\infty} a_n \quad \text{converges} \quad \Longleftrightarrow \quad \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges.}$$

Substituting $a_n = f(n)$ yields the desired result:

$$\sum_{n=1}^{\infty} f(n) \quad \text{converges} \quad \Longleftrightarrow \quad \sum_{k=0}^{\infty} 2^k f(2^k) \text{ converges.}$$

## Exercise 6.6

Show that

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$$

converges iff $p > 1$.

*Solution.* Define $a_n = \frac{1}{n(\log n)^p}$. For $n \geq 3$, $(a_n)$ is positive and decreasing. By the condensation test,

$$\sum_{n=2}^{\infty} a_n \quad \text{converges} \quad \Longleftrightarrow \quad \sum_{k=1}^{\infty} 2^k a_{2^k} \text{ converges.}$$

Now

$$2^k a_{2^k} = \frac{2^k}{2^k (\log 2^k)^p} = \frac{1}{(k \log 2)^p}.$$

Hence the condensed series is

$$\sum_{k=1}^{\infty} \frac{1}{(k \log 2)^p} = \frac{1}{(\log 2)^p} \sum_{k=1}^{\infty} \frac{1}{k^p}.$$

The $p$-series $\sum 1/k^p$ converges iff $p > 1$. Therefore the given series converges iff $p > 1$.

## Exercise 6.1

Suppose $f : [0, 1) \to [0, 1)$ is nonnegative, continuous, and monotone decreasing. Let $N, M \in \mathbb{Z}$.

a) Show that

$$\sum_{n=N}^{M} f(n) \;\geq\; \int_N^{M+1} f(x)\, dx.$$

b) Show that

$$\sum_{n=N+1}^{M} f(n) \;\leq\; \int_N^{M} f(x)\, dx.$$

c) Deduce

$$\int_N^{M+1} f(x)\, dx \;\leq\; \sum_{n=N}^{M} f(n) \;\leq\; f(N) + \int_N^{M} f(x)\, dx.$$

d) Show that $\sum_{n=N}^{\infty} f(n)$ converges iff the improper integral $\int_N^\infty f(x)\,dx$ exists.

e) Conclude that $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

*Solution.* (a) On $[n, n+1]$, $f(x) \le f(n)$, so

$$\int_n^{n+1} f(x)\,dx \le f(n).$$

Summing yields

$$\int_N^{M+1} f(x)\,dx \le \sum_{n=N}^{M} f(n).$$

(b) On $[n, n+1]$, $f(x) \ge f(n+1)$, so

$$\int_n^{n+1} f(x)\,dx \ge f(n+1).$$

Summing gives

$$\int_N^{M} f(x)\,dx \ge \sum_{n=N+1}^{M} f(n).$$

(c) Combining (a) and (b):

$$\int_N^{M+1} f(x)\,dx \le \sum_{n=N}^{M} f(n) \le f(N) + \int_N^{M} f(x)\,dx.$$

(d) The error term between the sum and integral is at most $f(N)$, independent of $M$. Hence $\sum f(n)$ converges iff $\int f(x)$ converges.

(e) For $f(x) = 1/x$, $\int_1^M \frac{1}{x}\,dx = \log M \to \infty$. So the harmonic series diverges.

## Exercise 6.2

Show that

$$\int_1^n \log x\,dx \ \le \ \sum_{k=1}^{n} \log k \ \le \ \int_1^n \log x\,dx + \log n.$$

Deduce that

$$\log(n!) \sim n\log n - n \quad (n \to \infty).$$

*Solution.* On each interval $[k, k+1]$, we have

$$\int_k^{k+1} \log x\,dx \ \le \ \log(k+1).$$

Summing over $k = 1, \ldots, n-1$ gives

$$\int_1^n \log x\,dx \ \le \ \sum_{k=2}^{n} \log k.$$

Adding $\log 1 = 0$ gives

$$\int_1^n \log x\,dx \ \le \ \sum_{k=1}^{n} \log k.$$

Similarly, one checks that

$$\sum_{k=1}^{n} \log k \;\le\; \int_1^n \log x\,dx + \log n.$$

Thus

$$n\log n - n + 1 \;\le\; \log(n!) \;\le\; n\log n - n + 1 + \log n.$$

Dividing through by $n\log n - n$ and letting $n \to \infty$ shows

$$\frac{\log(n!)}{n\log n - n} \to 1,$$

i.e.

$$\log(n!) \sim n\log n - n.$$

### Exercise 6.3

Let $f : [a,b] \to \mathbb{R}^m$ be bounded.

a) Define what it means for $f$ to be Riemann integrable.

b) Show that if each coordinate function $f_i : [a,b] \to \mathbb{R}$ is Riemann integrable, then $f$ is integrable and

$$\int_a^b f(x)\,dx = \left(\int_a^b f_1(x)\,dx,\ \dots,\ \int_a^b f_m(x)\,dx\right).$$

c) Conversely, show that if $f$ is integrable then each coordinate $f_i$ is integrable and the same equality holds.

*Solution.* (a) $f$ is Riemann integrable if there exists $I \in \mathbb{R}^m$ such that for every sequence of tagged partitions with mesh $\to 0$, $R(f, P_n, \tau_n) \to I$.

(b) Writing $f = (f_1, \dots, f_m)$, we have

$$R(f, P, \tau) = \big(R(f_1, P, \tau), \dots, R(f_m, P, \tau)\big).$$

If each $f_i$ is integrable, then each component converges to $\int f_i$. Hence $f$ is integrable and

$$\int_a^b f(x)\,dx = \left(\int_a^b f_1(x)\,dx,\ \dots,\ \int_a^b f_m(x)\,dx\right).$$

(c) Conversely, if $f$ is integrable with integral $I = (I_1, \dots, I_m)$, then $R(f, P_n, \tau_n) \to I$ implies $R(f_i, P_n, \tau_n) \to I_i$. Thus each $f_i$ is integrable and $\int f_i = I_i$.

### Exercise 6.4

Suppose $f, g : [a,b] \to \mathbb{R}^m$ are Riemann integrable and $\alpha, \beta \in \mathbb{R}$.

a) Show that $\alpha f + \beta g$ is integrable and

$$\int_a^b (\alpha f(x) + \beta g(x))\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx.$$

b) Show that

$$\left\| \int_a^b f(x)\,dx \right\| \le \int_a^b \|f(x)\|\,dx.$$

*Solution.* (a) For a tagged partition $(P, \tau)$,

$$R(\alpha f + \beta g, P, \tau) = \alpha R(f, P, \tau) + \beta R(g, P, \tau).$$

As $\operatorname{mesh}(P) \to 0$, these sums converge to the corresponding integrals. Hence

$$\int_a^b (\alpha f + \beta g) = \alpha \int_a^b f + \beta \int_a^b g.$$

(b) Again, for $(P, \tau)$,

$$\|R(f, P, \tau)\| = \left\| \sum f(t_j)(x_j - x_{j-1}) \right\| \leq \sum \|f(t_j)\|(x_j - x_{j-1}) = R(\|f\|, P, \tau).$$

Passing to the limit as $\operatorname{mesh}(P) \to 0$ gives

$$\left\| \int_a^b f(x)\,dx \right\| \leq \int_a^b \|f(x)\|\,dx.$$

## Exercise 6.5 (Optional)

Let $f, g : [a, b] \to \mathbb{R}^m$ be Riemann integrable.

a) Define the inner product

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x)\,dx,$$

and the corresponding norm

$$\|f\|_2 = \Big( \int_a^b \|f(x)\|^2\,dx \Big)^{1/2}.$$

b) Show the Cauchy–Schwarz inequality:

$$|\langle f, g \rangle| \leq \|f\|_2 \, \|g\|_2.$$

c) Deduce the Minkowski inequality:

$$\|f + g\|_2 \leq \|f\|_2 + \|g\|_2.$$

*Solution.* (a) The inner product is defined by

$$\langle f, g \rangle = \int_a^b f(x) \cdot g(x)\,dx,$$

with induced norm

$$\|f\|_2 = \big( \langle f, f \rangle \big)^{1/2} = \Big( \int_a^b \|f(x)\|^2\,dx \Big)^{1/2}.$$

(b) Pointwise inequality: $|f(x) \cdot g(x)| \leq \|f(x)\|\|g(x)\|$. Integrating gives

$$|\langle f, g \rangle| \leq \int_a^b \|f(x)\|\|g(x)\|\,dx.$$

Applying Hölder ($p = q = 2$) yields

$$\int_a^b \|f(x)\|\|g(x)\| \leq \|f\|_2\|g\|_2.$$

Thus

$$|\langle f, g\rangle| \leq \|f\|_2\|g\|_2.$$

(c) Expanding,

$$\|f + g\|_2^2 = \|f\|_2^2 + 2\langle f, g\rangle + \|g\|_2^2.$$

By (b), $2\langle f, g\rangle \leq 2\|f\|_2\|g\|_2$, so

$$\|f + g\|_2^2 \leq (\|f\|_2 + \|g\|_2)^2.$$

Taking square roots gives Minkowski:

$$\|f + g\|_2 \leq \|f\|_2 + \|g\|_2.$$

## Exercise 6.3

Suppose $f : [a, b] \to \mathbb{R}^m$ is integrable.

a) Show that if $x, y \in [a, b]$:

$$|\|f(x)\| - \|f(y)\|| \leq \sum_{l=1}^{m} |f^l(x) - f^l(y)|.$$

b) Deduce that if $I$ is any subinterval of $[a, b]$:

$$\sup_{x\in I} \|f(x)\| - \inf_{y\in I} \|f(y)\| \leq \sum_{l=1}^{m} \Big(\sup_{x\in I} f^l(x) - \inf_{y\in I} f^l(y)\Big).$$

c) Prove that $\|f\| : [a, b] \to \mathbb{R}$ is integrable.

*Solution.* (a) By the general inequality $|\|u\| - \|v\|| \leq \|u - v\|$,

$$|\|f(x)\| - \|f(y)\|| \leq \|f(x) - f(y)\|.$$

But

$$\|f(x) - f(y)\| \leq \sum_{l=1}^{m} |f^l(x) - f^l(y)|.$$

Hence the claim.

(b) Taking supremum over $x \in I$ and infimum over $y \in I$,

$$\sup_{x\in I} \|f(x)\| - \inf_{y\in I} \|f(y)\| \leq \sum_{l=1}^{m} \big(\sup_{x\in I} f^l(x) - \inf_{y\in I} f^l(y)\big).$$

(c) For any partition $P = (x_0, \ldots, x_k)$,

$$U(\|f\|, P) - L(\|f\|, P) = \sum_{j=1}^{k} \big(\sup_{x\in I_j} \|f(x)\| - \inf_{y\in I_j} \|f(y)\|\big)\Delta x_j.$$

By (b), this is at most

$$\sum_{l=1}^{m} \big(U(f^l, P) - L(f^l, P)\big).$$

Since each $f^l$ is integrable, the right-hand side can be made $< \varepsilon$ for sufficiently fine partitions. Hence $\|f\|$ is integrable.

### Exercise 6.4

Let $f : [a, b] \to \mathbb{R}^m$ be integrable, and write

$$f(t) = e_1 f^1(t) + e_2 f^2(t) + \cdots + e_m f^m(t),$$

for coordinate functions $f^i : [a, b] \to \mathbb{R}$.

a) For a tagged partition $(P, \tau)$, show that

$$R(f, P, \tau) = e_1 R(f^1, P, \tau) + \cdots + e_m R(f^m, P, \tau).$$

b) Show that if $\{(P_\ell, \tau_\ell)\}$ is a sequence of tagged partitions with mesh $\|P_\ell\| \to 0$, then

$$R(f, P_\ell, \tau_\ell) \to \int_a^b f(x)\, dx.$$

c) Show that for a tagged partition $(P, \tau)$,

$$\|R(f, P, \tau)\| \le R(\|f\|, P, \tau).$$

d) Deduce that

$$\left\| \int_a^b f(t)\, dt \right\| \le \int_a^b \|f(t)\|\, dt.$$

*Solution.* (a) Writing $f(t_j) = (f^1(t_j), \ldots, f^m(t_j))$, we have

$$R(f, P, \tau) = \sum_{j=1}^{k} f(t_j)\Delta x_j = \sum_{j=1}^{k}\sum_{i=1}^{m} e_i f^i(t_j)\Delta x_j = \sum_{i=1}^{m} e_i R(f^i, P, \tau).$$

(b) Since each $f^i$ is integrable, $R(f^i, P_\ell, \tau_\ell) \to \int_a^b f^i(x)\, dx$. Thus

$$R(f, P_\ell, \tau_\ell) \to \sum_{i=1}^{m} e_i \int_a^b f^i(x)\, dx = \int_a^b f(x)\, dx.$$

(c) By the triangle inequality,

$$\|R(f, P, \tau)\| = \left\| \sum_{j=1}^{k} f(t_j)\Delta x_j \right\| \le \sum_{j=1}^{k} \|f(t_j)\|\Delta x_j = R(\|f\|, P, \tau).$$

(d) For a refining sequence of partitions, (c) gives

$$\|R(f, P_\ell, \tau_\ell)\| \le R(\|f\|, P_\ell, \tau_\ell).$$

Taking limits as $\ell \to \infty$ yields

$$\left\| \int_a^b f(t)\, dt \right\| \le \int_a^b \|f(t)\|\, dt.$$

**Definition (Tagged partition).** A *partition* of $[a, b]$ is a finite ordered set

$$P = \{x_0, x_1, \dots, x_k\}, \quad a = x_0 < x_1 < \cdots < x_k = b.$$

Its *mesh* is

$$\|P\| = \max_{1 \le j \le k} (x_j - x_{j-1}).$$

A *tagged partition* is a partition $P = (x_0, \dots, x_k)$ together with points (tags) $t_j \in [x_{j-1}, x_j]$ for $j = 1, \dots, k$.

For a bounded function $f : [a, b] \to \mathbb{R}^m$, the *Riemann sum* with respect to $(P, \tau)$ is

$$R(f, P, \tau) = \sum_{j=1}^{k} f(t_j)\,(x_j - x_{j-1}).$$

## Exercise 6.5 (*)

Suppose $f, g, h : [a, b] \to \mathbb{R}^m$ are integrable. Define

$$(f, g) := \int_a^b \langle f(t), g(t) \rangle \, dt, \qquad \|f\|_{L^2} = \sqrt{(f, f)}.$$

a) Show that

$$(f, g) = (g, f), \quad (f + h, g) = (f, g) + (h, g), \quad (\lambda f, g) = \lambda (f, g).$$

b) For $t \in \mathbb{R}$, show that

$$\|f + tg\|_{L^2}^2 = \|f\|_{L^2}^2 + 2t(f, g) + t^2 \|g\|_{L^2}^2 \ge 0.$$

c) Deduce the *Cauchy–Schwarz inequality*:

$$|(f, g)| \le \|f\|_{L^2} \, \|g\|_{L^2}.$$

When does equality hold?

d) Deduce the *Minkowski inequality*:

$$\|f + g\|_{L^2} \le \|f\|_{L^2} + \|g\|_{L^2}.$$

*Solution.* (a) Symmetry: $(f, g) = (g, f)$ since dot product is symmetric. Linearity follows from properties of the integral:

$$(f + h, g) = (f, g) + (h, g), \qquad (\lambda f, g) = \lambda (f, g).$$

(b) Expanding,

$$\|f + tg\|_{L^2}^2 = (f + tg, f + tg) = (f, f) + 2t(f, g) + t^2 (g, g) = \|f\|_{L^2}^2 + 2t(f, g) + t^2 \|g\|_{L^2}^2 \ge 0.$$

(c) Since the quadratic in $t$ is nonnegative, its discriminant is nonpositive:

$$(2(f, g))^2 - 4\|f\|_{L^2}^2 \|g\|_{L^2}^2 \le 0.$$

Hence

$$|(f, g)| \leq \|f\|_{L^2}\|g\|_{L^2}.$$

Equality holds when $f$ and $g$ are linearly dependent almost everywhere.

(d) Expanding,

$$\|f+g\|_{L^2}^2 = \|f\|_{L^2}^2 + 2(f, g) + \|g\|_{L^2}^2.$$

By (c),

$$2(f, g) \leq 2\|f\|_{L^2}\|g\|_{L^2}.$$

Thus

$$\|f+g\|_{L^2}^2 \leq (\|f\|_{L^2} + \|g\|_{L^2})^2,$$

so

$$\|f+g\|_{L^2} \leq \|f\|_{L^2} + \|g\|_{L^2}.$$

## Exercise 7.1

Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ is given by $f(x) = x$.

*Solution.* For any $p \in \mathbb{R}^n$,

$$f(p+h) - f(p) = h.$$

Taking $L = \iota$ (the identity map),

$$\frac{\|f(p+h) - f(p) - Lh\|}{\|h\|} = \frac{\|h - \iota h\|}{\|h\|} = 0.$$

Hence $f$ is differentiable at every $p$ with derivative $Df(p) = \iota$.

## Exercise 7.2

Show that the map $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f\begin{pmatrix} x \\ y \end{pmatrix} = x^2 + y^2$$

is differentiable at all $p = (\xi, \eta) \in \mathbb{R}^2$ with Jacobian

$$Df(p) = (2\xi \;\; 2\eta).$$

*Solution.* The partial derivatives are

$$\frac{\partial f}{\partial x}(x, y) = 2x, \qquad \frac{\partial f}{\partial y}(x, y) = 2y.$$

So the gradient row vector is

$$Df(x, y) = (2x \;\; 2y).$$

At $p = (\xi, \eta)$,

$$Df(p) = (2\xi \;\; 2\eta).$$

Since $f$ is a polynomial, it is smooth and thus differentiable everywhere.

## Exercise 7.3

One might hope to compute the differential via

$$\lim_{x \to p} \frac{f(x) - f(p)}{\|x - p\|}.$$

Show this limit need not exist, even if $f$ is differentiable.

*Solution.* Take $f(x) = x : \mathbb{R}^n \to \mathbb{R}^n$, differentiable everywhere with derivative the identity. At $p = 0$,

$$\frac{f(x) - f(0)}{\|x\|} = \frac{x}{\|x\|}.$$

As $x \to 0$, this depends on the direction: along $e_1$ it tends to $e_1$, along $e_2$ to $e_2$. Thus the limit does not exist, though $f$ is differentiable at 0.

## Exercise 7.4

Suppose $\Omega \subset \mathbb{R}^n$ is open and $f, g : \Omega \to \mathbb{R}^m$ are differentiable at $p \in \Omega$. Show that $h = f + g$ is differentiable at $p$ and

$$Dh(p) = Df(p) + Dg(p).$$

*Solution.* For $h = f + g$,

$$h(p + h) - h(p) = (f(p + h) - f(p)) + (g(p + h) - g(p)).$$

Subtract $(Df(p) + Dg(p))h$ and divide by $\|h\|$. Each term tends to 0 by differentiability of $f$ and $g$. Hence $h$ is differentiable and $Dh(p) = Df(p) + Dg(p)$.

## Exercise 7.5

Suppose $\Omega, \Omega' \subset \mathbb{R}^n$ are open, $g : \Omega \to \Omega'$, $f : \Omega' \to \Omega$ with $f \circ g = \mathrm{id}_\Omega$, $g \circ f = \mathrm{id}_{\Omega'}$. Assume $g$ is differentiable at $p \in \Omega$ and $f$ is differentiable at $g(p) \in \Omega'$. Show

$$Df(g(p)) = (Dg(p))^{-1}.$$

*Solution.* From $f \circ g = \mathrm{id}_\Omega$, the chain rule at $p$ gives

$$Df(g(p)) \cdot Dg(p) = I.$$

From $g \circ f = \mathrm{id}_{\Omega'}$, differentiating at $g(p)$ gives

$$Dg(p) \cdot Df(g(p)) = I.$$

Thus $Df(g(p))$ is both a left- and right-inverse of $Dg(p)$, hence

$$Df(g(p)) = (Dg(p))^{-1}.$$

**Exercise 7.6 (*)**

(a) Show that the map $P : \mathbb{R}^2 \to \mathbb{R}$ given by

$$P\begin{pmatrix} x \\ y \end{pmatrix} = xy$$

is differentiable at $p = (\xi, \eta)$ with Jacobian

$$DP(p) = (\eta \;\; \xi).$$

*Solution.* Partial derivatives:

$$\frac{\partial P}{\partial x}(x, y) = y, \qquad \frac{\partial P}{\partial y}(x, y) = x.$$

At $(\xi, \eta)$,

$$DP(\xi, \eta) = (\eta \;\; \xi).$$

Since $P$ is a polynomial, it is differentiable everywhere.

(b) Suppose $f, g : \mathbb{R}^n \to \mathbb{R}$ are differentiable at $q$. Show that

$$Q(z) = \begin{pmatrix} f(z) \\ g(z) \end{pmatrix}$$

is differentiable with

$$DQ(q) = \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix}.$$

*Solution.* $Q$ has components $Q_1 = f$, $Q_2 = g$. Since $f, g$ are differentiable at $q$, $Q$ is differentiable with Jacobian

$$DQ(q) = \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix}.$$

(c) Show that $F(z) = f(z)g(z)$ is differentiable with

$$DF(q) = g(q)Df(q) + f(q)Dg(q).$$

*Solution.* Write $F = P \circ Q$, where $P(x, y) = xy$, $Q(z) = (f(z), g(z))$. By the chain rule,

$$DF(q) = DP(Q(q)) \cdot DQ(q).$$

Now $DP(Q(q)) = (g(q) \;\; f(q))$, and $DQ(q) = \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix}$. So

$$DF(q) = (g(q) \;\; f(q)) \begin{pmatrix} Df(q) \\ Dg(q) \end{pmatrix} = g(q)Df(q) + f(q)Dg(q).$$

### Exercise 8.1

Show that $f : \mathbb{R}^2 \to \mathbb{R}$ is everywhere differentiable and compute the differential.

a) $f(x, y) = x^2 + y^2 - x - xy$.

$$Df(x, y) = (2x - 1 - y,\ 2y - x).$$

b) $f(x, y) = \dfrac{1}{\sqrt{1 + x^2 + y^2}}$.

$$Df(x, y) = \left( -\frac{x}{(1 + x^2 + y^2)^{3/2}},\ -\frac{y}{(1 + x^2 + y^2)^{3/2}} \right).$$

c) $f(x, y) = x^5 y^2$.

$$Df(x, y) = (5x^4 y^2,\ 2x^5 y).$$

### Exercise 8.2

Find the minimum of

$$f(x, y, z) = x^4(y^2 + x^2) + z^2 - 4z.$$

*Solution.* $f_x = 2x^3(2y^2 + 3x^2)$, $f_y = 2x^4 y$, $f_z = 2z - 4$. Solving $\nabla f = 0$ gives $x = 0, z = 2$, $y$ arbitrary.

Thus the critical set is $\{(0, y, 2) : y \in \mathbb{R}\}$. At these points,

$$f(0, y, 2) = -4.$$

Hence the minimum value is $-4$, attained along the line $\{(0, y, 2) : y \in \mathbb{R}\}$.

### Exercise 8.3

Suppose $A$ is a symmetric $n \times n$ matrix and $f(x) = x^t A x$.

a) Show that $f$ is differentiable with

$$Df(p) = 2p^t A.$$

*Solution.* $\nabla f(x) = (A + A^t)x = 2Ax$ since $A$ is symmetric. Thus $Df(p) = (2Ap)^t = 2p^t A$.

b) Find Hess $f(p)$.

*Solution.* $\nabla f(x) = 2Ax$, so the derivative is constant:

$$\operatorname{Hess} f(p) = 2A.$$

### Exercise 8.4 (*)

Consider

$$f(x,y)=\begin{cases}\dfrac{xy^3-x^3y}{x^2+y^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0).\end{cases}$$

a) Show that

$$D_1f(x,y)=\begin{cases}\dfrac{y^3-3x^2y}{x^2+y^2}-\dfrac{2x(xy^3-x^3y)}{(x^2+y^2)^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0),\end{cases}$$

and

$$D_2f(x,y)=\begin{cases}\dfrac{3y^2x-x^3}{x^2+y^2}-\dfrac{2y(xy^3-x^3y)}{(x^2+y^2)^2}, & (x,y)\neq(0,0),\\ 0, & (x,y)=(0,0).\end{cases}$$

b) Show that $D_1f$ and $D_2f$ are continuous at $(0,0)$.

*Solution.* For $(x,y)\neq(0,0)$ the formulas follow by quotient rule. At $(0,0)$ the limits defining partial derivatives equal 0. In polar coordinates $(x,y)=(r\cos\theta,r\sin\theta)$, both $D_1f,D_2f$ are $O(r)$ as $r\to 0$, hence tend to 0. Thus both are continuous at $(0,0)$.

### Exercise 8.4 (b,c)

b) Along $te_2=(0,t)$,

$$D_1f(0,t)=t \;\Rightarrow\; \frac{D_1f(0,t)-D_1f(0,0)}{t}=1.$$

Along $te_1=(t,0)$,

$$D_2f(t,0)=-t \;\Rightarrow\; \frac{D_2f(t,0)-D_2f(0,0)}{t}=-1.$$

c) Hence

$$D_2D_1f(0)=1,\quad D_1D_2f(0)=-1.$$

So both exist, but are not equal.

### Exercise 8.5 (*)

a) By induction on $m$, prove the multinomial theorem:

$$(x_1+\cdots+x_m)^j=\sum_{n_1+\cdots+n_m=j}\binom{j}{n_1,\ldots,n_m}x_1^{n_1}\cdots x_m^{n_m}.$$

*Solution.* For $m=1$ trivial; for $m=2$ this is the binomial theorem. Inductive step: write

$$(x_1+\cdots+x_m)^j=\sum_{k=0}^{j}\binom{j}{k}(x_1+\cdots+x_{m-1})^{j-k}x_m^k.$$

Apply induction to $(x_1+\cdots+x_{m-1})^{j-k}$, relabel $n_m=k$, and simplify coefficients to obtain

$$\binom{j}{n_1,\ldots,n_m}=\frac{j!}{n_1!\cdots n_m!}.$$

b) Expanding

$$(x_1 + \cdots + x_m)^j = (x_1 + \cdots + x_m)(x_1 + \cdots + x_m)^{j-1},$$

the coefficient of $x_1^{n_1} \cdots x_m^{n_m}$ on the LHS is $\binom{j}{n_1,\ldots,n_m}$. On the RHS it comes from choosing $x_i$ from the first factor, giving

$$\binom{j-1}{n_1, \ldots, n_i - 1, \ldots, n_m}.$$

Summing over $i = 1, \ldots, m$ gives the recursive identity.

## Exercise 8.6

Let $\Omega = \{(x, y)^t \in \mathbb{R}^2 : x > 0\}$ and

$$f(x, y) = \begin{pmatrix} x \sin y \\ x \cos y \end{pmatrix}.$$

a) Compute derivatives:

$$Df(\xi, \eta) = \begin{pmatrix} \sin \eta & \xi \cos \eta \\ \cos \eta & -\xi \sin \eta \end{pmatrix}.$$

Thus $f$ is differentiable everywhere.

b) $\det Df(\xi, \eta) = -\xi \neq 0$ for $(\xi, \eta) \in \Omega$, hence invertible.

c) $f$ is not injective since $f(x, y) = f(x, y + 2\pi)$. So invertibility of the derivative does not imply global injectivity. This illustrates the local nature of the Inverse Function Theorem.

## Exercise 8.7

a) Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuously differentiable near 0 and $f'(0) = 0$. We need an example showing that $f$ may still be bijective.

*Solution.* Consider

$$f(x) = x^3.$$

Then $f'(x) = 3x^2$, so $f'(0) = 0$. Nevertheless, $f$ is strictly increasing on $\mathbb{R}$, hence bijective with inverse

$$f^{-1}(y) = \sqrt[3]{y}.$$

Thus, even if the derivative at the origin vanishes, $f$ may still be bijective in one dimension.

b) Suppose $f : \mathbb{R}^n \to \mathbb{R}^n$ is bijective, differentiable at the origin, and $\det Df(0) = 0$. Show that $f^{-1}$ is not differentiable at $f(0)$.

*Solution.* Assume for contradiction that $f^{-1}$ is differentiable at $f(0)$. Define the identity map $\iota : \mathbb{R}^n \to \mathbb{R}^n$ by $\iota(x) = x$. Note that $\iota = f^{-1} \circ f$.

Differentiating at 0 and applying the chain rule gives

$$D\iota(0) = D(f^{-1})(f(0)) \cdot Df(0).$$

But $D\iota(0) = I_n$, the identity matrix. Thus

$$I_n = D(f^{-1})(f(0)) \cdot Df(0).$$

This equation says that $Df(0)$ must have a right inverse, which would imply that $Df(0)$ is invertible. But $\det Df(0) = 0$, so $Df(0)$ is singular and not invertible. This is a contradiction.

Therefore, $f^{-1}$ cannot be differentiable at $f(0)$.

## Exercise 9.1

Let $\gamma : [0, 2\pi] \to \mathbb{R}^2$ be given by

$$\gamma(\theta) = \begin{pmatrix} \theta - \sin\theta \\ 1 - \cos\theta \end{pmatrix}.$$

**Definition (Parametrised curve).** A *parametrised curve* in $\mathbb{R}^2$ is a map $\gamma : [a, b] \to \mathbb{R}^2$. We say $\gamma$ is of class $C^1$ if its coordinate functions are continuously differentiable.

**Definition (Arc length).** For a $C^1$ curve $\gamma : [a, b] \to \mathbb{R}^2$, the *arc length* is

$$L(\gamma) = \int_a^b \|\gamma'(t)\| \, dt,$$

where $\|\cdot\|$ is the Euclidean norm in $\mathbb{R}^2$.

**Definition (Cycloid).** The *cycloid* is the path traced by a fixed point on the circumference of a circle of radius 1 rolling along the $x$-axis without slipping. Its standard parametrisation is

$$\gamma(\theta) = (\theta - \sin\theta,\ 1 - \cos\theta), \quad 0 \le \theta \le 2\pi.$$

a) *Sketch.*

At $\theta = 0$, $\gamma(0) = (0, 0)$. At $\theta = \pi$, $\gamma(\pi) = (\pi, 2)$. At $\theta = 2\pi$, $\gamma(2\pi) = (2\pi, 0)$.

Hence $\gamma$ describes a single arch of the cycloid: it starts at the origin, rises to height 2 above $x = \pi$, and returns to the $x$-axis at $(2\pi, 0)$. It is symmetric about the vertical line $x = \pi$.

b) *Arc length.*

Differentiate:

$$\gamma'(\theta) = (1 - \cos\theta,\ \sin\theta).$$

Compute norm:

$$\|\gamma'(\theta)\|^2 = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta = 4\sin^2\!\left(\frac{\theta}{2}\right).$$

Thus

$$\|\gamma'(\theta)\| = 2\sin\!\left(\frac{\theta}{2}\right), \quad 0 \le \theta \le 2\pi.$$

Hence

$$L(\gamma) = \int_0^{2\pi} 2\sin\!\left(\frac{\theta}{2}\right) d\theta.$$

Substitute $u = \frac{\theta}{2}$, so $d\theta = 2du$, limits $0 \mapsto 0$, $2\pi \mapsto \pi$:

$$L(\gamma) = 4\int_0^{\pi} \sin u \, du = 4[-\cos u]_0^{\pi} = 4(1 - (-1)) = 8.$$

$$\boxed{L(\gamma) = 8}$$

## Exercise 9.2

Let $A = [a_1, b_1] \times [a_2, b_2] \subset \mathbb{R}^2$ be a rectangle, and let $\mathcal{P}$ be a partition of $A$. Suppose $f, g : A \to \mathbb{R}$ are bounded and $\lambda \ge 0$.

**Definition (Upper and Lower Sums).** For a bounded function $f : A \to \mathbb{R}$ and a partition $\mathcal{P}$ of $A$ into rectangles $R$, the *upper sum* is

$$U(f,\mathcal{P}) = \sum_{R\in\mathcal{P}} \big(\sup_R f\big)\,|R|,$$

and the *lower sum* is

$$L(f,\mathcal{P}) = \sum_{R\in\mathcal{P}} \big(\inf_R f\big)\,|R|,$$

where $|R|$ denotes the area of the rectangle $R$.

**Background.** These constructions satisfy simple algebraic rules: - taking negatives exchanges suprema and infima, - scaling by a positive constant factors out, - suprema are subadditive, infima are superadditive.

a) $U(-f,\mathcal{P}) = -L(f,\mathcal{P})$, since $\sup_R(-f) = -\inf_R f$.
b) $L(-f,\mathcal{P}) = -U(f,\mathcal{P})$, since $\inf_R(-f) = -\sup_R f$.
c) $U(\lambda f,\mathcal{P}) = \lambda U(f,\mathcal{P})$, since $\sup_R(\lambda f) = \lambda \sup_R f$.
d) $L(\lambda f,\mathcal{P}) = \lambda L(f,\mathcal{P})$, since $\inf_R(\lambda f) = \lambda \inf_R f$.
e) $U(f+g,\mathcal{P}) \le U(f,\mathcal{P}) + U(g,\mathcal{P})$, since $\sup_R(f+g) \le \sup_R f + \sup_R g$.
f) $L(f+g,\mathcal{P}) \ge L(f,\mathcal{P}) + L(g,\mathcal{P})$, since $\inf_R(f+g) \ge \inf_R f + \inf_R g$.

Hence the algebraic properties of upper and lower sums hold as claimed.

## Exercise 9.3

Let $A = [-1,1]\times[-1,1]$ and define

$$f(x,y) = \begin{cases} 1, & x \notin \mathbb{Q},\ y = 0, \\ 0, & \text{otherwise.} \end{cases}$$

**Background.** - The function $f$ takes values only 0 or 1, so it is bounded. - It is nonzero only along the horizontal line segment

$$S = \{(x,0) : x \in [-1,1],\ x \notin \mathbb{Q}\}.$$

- The set $S$ is a subset of a line in $\mathbb{R}^2$, and such sets have *measure zero.* - For the Riemann integral, bounded functions supported on measure-zero sets are integrable with integral 0.

**Solution.** Since $S$ has measure zero, the contribution of $f$ to any upper sum can be made arbitrarily small. Thus $f$ is Riemann integrable on $A$, and

$$\int_A f(z)\,dz = 0.$$

**Conclusion.** Although $f$ takes the value 1 on uncountably many points (all irrationals on the $x$-axis between $-1$ and 1), this set is "too thin" to contribute to the integral. The integral only depends on two-dimensional area, not one-dimensional subsets.

## Exercise 9.4

Let $A = [a_1,b_1]\times[a_2,b_2]$ and $|A| = (b_1-a_1)(b_2-a_2)$. Suppose $f,g : A \to \mathbb{R}$ are integrable.

**Background.** For a bounded function $f$ on $A$, let

$$m = \inf_{x\in A} f(x), \quad M = \sup_{x\in A} f(x).$$

Then $m \le f(x) \le M$ for all $x \in A$. Multiplying by the area $|A|$ and integrating yields inequalities for the integral. Key facts:

- If $f \ge 0$, then $\int_A f \ge 0$.
- If $f \le g$, then $\int_A f \le \int_A g$.
- The triangle inequality: $\left|\int f\right| \le \int |f|$.

a) For all $x \in A$, $\inf_A f \le f(x) \le \sup_A f$. Integrating:

$$|A| \inf_{x\in A} f(x) \;\le\; \int_A f(x)\,dx \;\le\; |A| \sup_{x\in A} f(x).$$

b) Applying (a) to both $f$ and $-f$:

$$-|A| \sup_{x\in A} |f(x)| \le \int_A f(x)\,dx \le |A| \sup_{x\in A} |f(x)|.$$

Thus

$$\left|\int_A f(x)\,dx\right| \le |A| \sup_{x\in A} |f(x)|.$$

c) If $f(x) \ge 0$, then $\inf_A f \ge 0$. By (a),

$$0 \le \int_A f(x)\,dx.$$

d) If $f(x) \le g(x)$, then $0 \le g(x) - f(x)$. By (c),

$$0 \le \int_A (g-f) = \int_A g - \int_A f,$$

so

$$\int_A f \le \int_A g.$$

e) Since $-f(x) \le |f(x)|$ and $f(x) \le |f(x)|$, integrating gives

$$-\int_A |f| \le \int_A f \le \int_A |f|.$$

Thus

$$\left|\int_A f(x)\,dx\right| \le \int_A |f(x)|\,dx.$$

## Exercise 9.5

Let $A = [a_1, b_1] \times [a_2, b_2]$ and $f : A \to \mathbb{R}$ continuous. Define

$$F(y) = \int_{a_1}^{b_1} f(x,y)\,dx, \quad y \in [a_2, b_2].$$

**Background.** Since $f$ is continuous on the compact rectangle $A$, it is uniformly continuous. This allows us to control the difference $|f(x, y_1) - f(x, y_2)|$ uniformly in $x$.

*Solution.* Fix $y_1, y_2$. Then

$$|F(y_1) - F(y_2)| = \left| \int_{a_1}^{b_1} \big(f(x, y_1) - f(x, y_2)\big)\, dx \right| \leq \int_{a_1}^{b_1} |f(x, y_1) - f(x, y_2)|\, dx.$$

By uniform continuity, for $\varepsilon > 0$ there exists $\delta > 0$ such that if $|y_1 - y_2| < \delta$ then $|f(x, y_1) - f(x, y_2)| < \frac{\varepsilon}{b_1 - a_1}$ for all $x$. Hence

$$|F(y_1) - F(y_2)| \leq (b_1 - a_1) \tfrac{\varepsilon}{b_1 - a_1} = \varepsilon.$$

Thus $F$ is continuous.

## Exercise 9.6

a) The improper integral

$$\int_1^\infty e^{-u}\, du = [-e^{-u}]_1^\infty = e^{-1}$$

converges.

b) For $R > 1$,

$$\int_1^R e^{-u^2}\, du \leq \int_1^R e^{-u}\, du < \infty,$$

so $\int_1^\infty e^{-u^2}\, du$ converges. By symmetry the whole integral

$$I = \int_{-\infty}^\infty e^{-u^2}\, du$$

converges.

c) For $f \geq 0$, with $S_R = [-R, R] \times [-R, R]$ and $B_R(0)$ the disk,

$$\int_{B_R(0)} f \leq \int_{S_R} f \leq \int_{B_{\sqrt{2}R}(0)} f,$$

since $B_R(0) \subset S_R \subset B_{\sqrt{2}R}(0)$.

d) For $f(u, v) = e^{-(u^2+v^2)}$,

$$\iint_{\mathbb{R}^2} e^{-(u^2+v^2)}\, dudv = \left( \int_{-\infty}^\infty e^{-u^2}\, du \right)^2 = I^2.$$

In polar coordinates,

$$\iint_{\mathbb{R}^2} e^{-(u^2+v^2)}\, dudv = \int_0^{2\pi} \int_0^\infty e^{-r^2} r\, dr d\theta = 2\pi \cdot \tfrac{1}{2} \int_0^\infty e^{-t}\, dt = \pi.$$

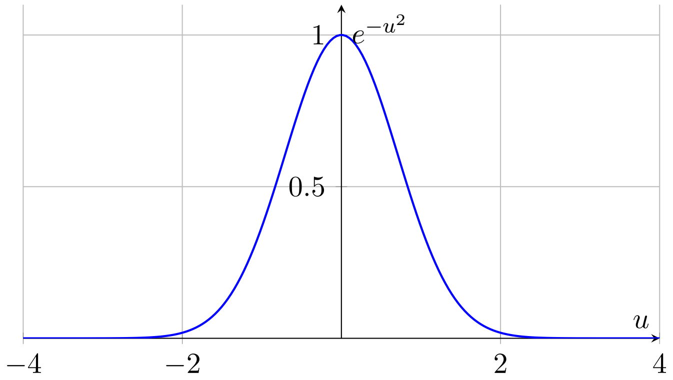
So $I^2 = \pi$.

e) Therefore

$$I = \sqrt{\pi}.$$

## Exercise 9.6 (with integrand chart)

The integrand is $f(u) = e^{-u^2}$. It is symmetric, maximised at $u = 0$ with $f(0) = 1$, and decays rapidly to 0 as $|u| \to \infty$. This rapid decay ensures convergence of the Gaussian integral.




This graph shows the bell-shaped curve of $e^{-u^2}$. It is clear that the function is bounded and rapidly decays to zero outside a neighbourhood of 0, so the area under the curve converges. Indeed,

$$I = \int_{-\infty}^{\infty} e^{-u^2}\,du = \sqrt{\pi}.$$

**Polar Coordinates.** To evaluate

$$I = \int_{-\infty}^{\infty} e^{-u^2}\,du,$$

we square it:

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-u^2}\,du\right)\left(\int_{-\infty}^{\infty} e^{-v^2}\,dv\right) = \iint_{\mathbb{R}^2} e^{-(u^2+v^2)}\,du\,dv.$$

Since the integrand depends only on $u^2 + v^2 = r^2$, we switch to polar coordinates:

$$u = r\cos\theta, \quad v = r\sin\theta, \qquad du\,dv = r\,dr\,d\theta.$$

Thus

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta = \left(\int_0^{2\pi} d\theta\right)\left(\int_0^{\infty} re^{-r^2}\,dr\right).$$

Let $t = r^2$, $dt = 2r\,dr$:

$$\int_0^{\infty} re^{-r^2}\,dr = \tfrac{1}{2}\int_0^{\infty} e^{-t}\,dt = \tfrac{1}{2}.$$

So

$$I^2 = 2\pi \cdot \tfrac{1}{2} = \pi, \qquad I = \sqrt{\pi}.$$